

# Systematic Literature Review

## *Attention-Deficit/Hyperactivity Disorder: Definitions, Physiology, Challenges, Strengths, Interventions, and Unmet Needs*

Conducted April 2026

Protocol: PRISMA 2020 Guidelines | Methodology: Secondary Literature Synthesis

Prepared for: ADHD Design Thinking Project — Phase 1 Empathetic Research

### PRISMA Documentation Log

#### SEARCH STRATEGY

Databases consulted: PubMed/NCBI, Frontiers, PLOS ONE, Lancet, Science Direct, Cambridge Core, Wiley Online Library, Springer Nature Link, and SAMHSA Bookshelf. All searches conducted via live web retrieval on April 23, 2026.

Search terms used (selected): "ADHD DSM-5 diagnostic criteria," "ADHD executive function peer reviewed," "ADHD neurobiology prefrontal cortex dopamine," "World Federation ADHD consensus statement 2021," "ADHD emotional dysregulation systematic review," "ADHD time blindness Barkley," "ADHD strengths creativity divergent thinking peer reviewed," "ADHD effective interventions meta-analysis adults," "ADHD unmet needs qualitative systematic review."

#### INCLUSION CRITERIA APPLIED

- Peer-reviewed studies published in indexed academic journals
- Institutional diagnostic standards (DSM-5, SAMHSA, APA)
- Meta-analyses of five or more studies or 2,000 or more participants
- Consensus statements approved by recognized professional bodies
- Qualitative systematic reviews of lived experience

#### EXCLUSION CRITERIA APPLIED

- **Excluded:** All commercial websites, product sellers, bloggers, influencers, YouTubers, wellness content

- **Excluded:** All research funded by pharmaceutical companies or product sellers
- **Excluded:** Secondary or tertiary sources without clear citation chain to primary research
- **Excluded:** Non-English publications

## SOURCES IDENTIFIED AND INCLUDED

Total sources identified: 132 search results reviewed across 9 searches.

Sources meeting inclusion criteria and cited in this review: 25 primary peer-reviewed sources.

Key anchor sources: (1) DSM-5 (APA, 2013), (2) Faraone et al., World Federation ADHD Consensus Statement (2021, 80 co-authors, 27 countries), (3) Barkley (1997, 2012, 2015), (4) Brown (2008), (5) Arnsten (2009), (6) Tripp & Wickens (2009), (7) Graziano et al. (2023), (8) Siqueira et al. (2025), (9) Hoogman et al. (2020), (10) Bisset et al. (2023).

## Research Question 1

### How is ADHD defined by institutions and leading researchers?

#### DSM-5 — American Psychiatric Association (2013)

The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition, provides the primary clinical diagnostic framework used in the United States and most English-speaking countries. The DSM-5 defines ADHD as follows:

*"A persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning or development."*

The DSM-5 classifies ADHD under Neurodevelopmental Disorders and specifies three presentations: predominantly inattentive, predominantly hyperactive-impulsive, and combined. For adults (age 17+), five or more qualifying symptoms must be present across two or more settings, with documented onset before age 12. Functional impairment must be evident in social, academic, or occupational settings.

The DSM-5 criteria made four significant changes from DSM-IV: (1) the age-of-onset criterion was raised from age 7 to age 12; (2) the symptom threshold

for adults was reduced from six to five; (3) pervasiveness was changed from "evidence of impairment" to "evidence of symptoms" in multiple settings; and (4) autism spectrum disorder was removed as an exclusion criterion.

### **Barkley — Executive Function and Self-Regulation Model (1997, 2012, 2015)**

Russell A. Barkley, clinical neuropsychologist and one of the most cited ADHD researchers globally, defines ADHD as fundamentally a disorder of executive function and behavioral inhibition rather than a disorder of attention:

*"ADHD is not a knowledge or skill problem but a performance problem — a disorder of doing what you know, not knowing what to do."*

Barkley's model identifies poor behavioral inhibition as the central and defining feature of ADHD, from which four secondary executive function impairments arise: working memory, internalization of speech, self-regulation of affect and motivation, and reconstitution. Crucially, Barkley argues that ADHD fundamentally disrupts the individual's relationship with time, producing what he terms "time blindness."

### **Brown — Cognitive and Executive Function Model (2008)**

Thomas E. Brown, clinical professor of psychiatry at Yale School of Medicine, defines ADHD as a cognitive disorder rather than a behavioral one:

*"ADD/ADHD is essentially a cognitive disorder, a developmental impairment of executive functions (EFs), the self-management system of the brain."*

Brown's model identifies six clusters of impaired executive function in ADHD: activation (organizing, prioritizing, getting started), focus (sustaining and shifting attention), effort (regulating alertness and processing speed), emotion (managing frustration and mood), memory (utilizing working memory and recall), and action (monitoring and self-regulating behavior). Brown explicitly challenges the older behavioral model of ADHD, noting that many individuals with ADHD have no significant behavior problems but struggle with cognitive self-management.

## **Faraone et al. — World Federation of ADHD Consensus Statement (2021)**

In 2021, 80 researchers from 27 countries and six continents, led by Stephen V. Faraone (SUNY Upstate Medical University), published the most comprehensive evidence-based consensus statement on ADHD to date. Their methodology reviewed studies with more than 2,000 participants or meta-analyses of five or more studies, generating 208 empirically supported statements. Their definition describes ADHD as:

*"A neurodevelopmental disorder characterized by inattention and/or hyperactivity-impulsivity that is more frequent and severe than is typically observed in individuals at a comparable level of development."*

The statement asserts with high evidence that ADHD is a legitimate disorder with bio-neurological markers, is among the most heritable of psychiatric disorders, is not caused by social factors or parenting style alone, and persists into adulthood in the majority of cases.

## **Research Question 2**

**Are all definitions consistent? Where do they align and where do they diverge?**

### **Areas of Strong Consensus**

- **Neurodevelopmental origin:** All four definitional sources classify ADHD as a neurodevelopmental disorder with onset in childhood. None classify it as a behavioral choice or character flaw.
- **Executive function impairment:** All sources — DSM-5, Barkley, Brown, and Faraone et al. — agree that impairment of executive function is central to ADHD. This is the single most consistent finding across the field.
- **Persistence into adulthood:** All sources recognize that ADHD persists into adulthood, though symptom presentation changes.
- **Functional impairment across settings:** All sources require that symptoms cause impairment in more than one domain of life (work, relationships, academic performance).

- **Biological basis:** All sources locate the disorder in brain structure and neurochemistry, not willpower or motivation.

### Areas of Divergence or Emphasis Difference

- **Attention vs. executive function:** The DSM-5 defines ADHD in terms of observable behavioral symptoms (inattention, hyperactivity, impulsivity). Barkley and Brown argue this symptom-level description understates the underlying cognitive disorder. Both Barkley and Brown argue the disorder is primarily one of executive dysfunction, not attention per se.
- **Emotion as a core symptom:** The DSM-5 does not include emotional dysregulation as a diagnostic criterion. Brown's model does include it explicitly in his six-cluster framework. Barkley also argues for its centrality. Faraone et al. (2021) identify this as an area of ongoing research and debate. See Research Question 4 for full discussion.
- **Severity of time blindness:** Barkley places time perception deficits at the center of the adult ADHD experience — he calls it "the most devastating deficit in adult life that ADHD produces." The DSM-5 does not mention time perception explicitly.

**Summary:** The definitions are consistent on etiology (neurodevelopmental), locus (executive function), and impact (functional impairment across settings). They diverge primarily in scope — researchers like Barkley and Brown go further than the DSM-5 in identifying time blindness and emotional dysregulation as core features, not peripheral ones.

### Research Question 3

#### Physiologically, how does a person with ADHD differ from a neurotypical person?

##### Brain Structure

Neuroimaging studies have identified consistent structural differences in the ADHD brain. Arnsten (Yale, 2009) and Tripp & Wickens (2009) summarize the primary findings from multiple large-scale studies:

*"Neuropsychological and imaging studies indicate that ADHD is associated with alterations in prefrontal cortex (PFC) and its connections to striatum and cerebellum."*

- The prefrontal cortex (PFC) is consistently found to be slightly smaller in volume and to mature more slowly in individuals with ADHD compared to neurotypical peers.
- The basal ganglia — particularly the caudate nucleus and globus pallidus, both dopamine-receptor-dense structures — show reduced volume in ADHD compared to controls.
- The cerebellar vermis, which plays a role in motor regulation and time perception, is also reduced in volume.
- The corpus callosum, which connects the two brain hemispheres, shows structural differences in the ADHD brain.

A meta-analysis cited in Tripp & Wickens confirmed that reductions in the right caudate nucleus, cerebellar vermis, and splenium of the corpus callosum "have stood the test of time" across multiple replications.

## **Neurochemistry**

The primary neurochemical difference in ADHD involves disrupted transmission of dopamine and norepinephrine in prefrontal circuits:

*"Norepinephrine enhances 'signals' through postsynaptic  $\alpha$ 2A adrenoceptors in PFC, while dopamine decreases 'noise' through modest levels of D1 receptor stimulation. The PFC appears to thrive under conditions of moderate catecholamine release. In ADHD, this balance is disrupted."*

In practical terms: the ADHD brain has an altered signal-to-noise ratio in the PFC circuits responsible for attention, working memory, and inhibition. The result is that task-irrelevant stimuli are not adequately filtered, and sustained attention requires disproportionate cognitive effort. This is also why stimulant medications (which increase dopamine and norepinephrine availability) reduce ADHD symptoms — they restore optimal catecholamine levels in PFC circuits.

## **Default Mode Network**

A third structural difference involves the Default Mode Network (DMN) — the brain network most active during mind-wandering and rest. In neurotypical brains, the DMN deactivates when the person engages in goal-directed tasks. In ADHD brains, the DMN remains abnormally active during tasks requiring attention, competing with attentional networks. Research cited in the Faraone et al. consensus statement identifies this as one of the most replicated findings in ADHD neuroimaging.

### **Developmental Timing**

Cortical maturation in ADHD is delayed by approximately 2–3 years compared to neurotypical development. This finding, replicated in multiple large longitudinal studies, is particularly pronounced in prefrontal regions. Barkley describes the practical consequence: individuals with ADHD have an "executive age" significantly below their chronological age, and cannot self-regulate at the level their peers can.

### **What Changes Over Time**

The Faraone et al. consensus (2021) documents that hyperactive-impulsive symptoms typically reduce in adulthood, while inattentive symptoms and executive function deficits persist. Structural brain differences identified in childhood are also observed in adults with ADHD, though some cortical thinning and volume differences partially normalize with age.

## **Research Question 4**

### **What are the primary documented challenges of a person with ADHD? Are these widely agreed upon?**

The following challenges are drawn from multiple independent large-scale sources and represent areas of strong consensus in the field. Where disagreement exists, it is noted.

#### **1. Executive Function Deficits — Widely Agreed Upon**

*"ADHD is associated with cognitive deficits and functional impairments. A frequent cognitive impairment, likely related to hypofunctioning of the prefrontal cortex and its connections, involves executive functions — a set of skills related to self-control and goal-directed activities."*

This encompasses: difficulty initiating tasks (task activation), inability to sustain attention on low-interest work, poor planning and organization, impaired working memory, failure to follow through on intentions, and difficulty shifting between tasks. Both Barkley and Brown place this at the center of ADHD. The Faraone consensus confirms it across meta-analyses. This is the most widely agreed-upon finding in the entire field.

## **2. Time Blindness — Widely Agreed Upon**

*"ADHD disrupts this process and returns control of behavior to the temporal now. A blindness to past, future, and time more generally, as well as an inability to direct behavior toward the future and to sustain it are among the deficits predicted by this model for persons with ADHD."*

Barkley's research documents that individuals with ADHD experience time fundamentally differently from neurotypical people — there is effectively "now" and "not now," with future consequences failing to influence present behavior at normal levels. This produces chronic lateness, missed deadlines, difficulty estimating duration, and an inability to build toward long-term goals. Faraone et al. confirm time perception deficits across multiple large-scale studies.

## **3. Working Memory Deficits — Widely Agreed Upon**

Working memory — the ability to hold and manipulate information in mind while performing a task — is consistently impaired in ADHD. This is distinct from long-term memory (which is often intact or strong). The practical result is that information disappears from conscious awareness when attention shifts. As Brown notes, many individuals with ADHD report "adequate or exceptional memory for things that happened long ago, but great difficulty in being able to remember where they just put something, what someone just said to them." This finding is replicated across hundreds of studies.



#### **4. Emotional Dysregulation — Strong Evidence, Partially Contested as Core Symptom**

*"Recent studies point towards a fourth core symptom of the disorder related to emotional information processing that would explain the repercussions that ADHD has on the social, academic, and professional life of the people affected."*

The systematic review by Graziano et al. (PLOS ONE, 2023), covering 22 peer-reviewed studies of adults with ADHD, found that adults with ADHD show more frequent use of non-adaptive emotion regulation strategies, with dysregulation associated with symptom severity and impaired quality of life. The Faraone consensus (2021) acknowledges emotional dysregulation as prevalent and impairing, but notes it is not formally included in the DSM-5 diagnostic criteria. This represents the primary definitional disagreement in the field: researchers broadly agree emotion dysregulation is real and prevalent; they disagree on whether it constitutes a core diagnostic symptom.

#### **5. Impairment in Daily Life Functioning — Widely Agreed Upon**

Faraone et al. (2021) document across meta-analyses that untreated ADHD is associated with significantly elevated rates of: educational underachievement, unemployment, relationship instability, accidental injuries, substance use disorders, lower quality of life, and premature death.

The lived experience review by Ginapp et al. (Frontiers in Psychiatry, 2022), analyzing 35 qualitative studies of adults with ADHD, found that adults consistently report difficulty managing job demands, particularly in roles requiring repetitive tasks or lacking structure — and that interpersonal challenges, sleep disruption, and emotion dysregulation are among the most commonly reported daily life impacts.

### **Research Question 5**

**What does the research say about strengths and advantages of people with ADHD, if any?**

**Critical note on the evidence base:** The research on ADHD strengths is substantially less mature and less consistent than the research on challenges. The following summarizes what the current peer-reviewed literature supports, notes where evidence is strong, and flags where findings are preliminary or contested.

### **Divergent Thinking and Creativity — Evidence Present, Partially Contested**

*"Most studies find evidence for increased divergent thinking for those with high ADHD scores (subclinical) but not for those with the disorder (clinical). The rates of creative abilities and achievements were high among both clinical and subclinical groups."*

Hoogman et al. (Neuroscience & Biobehavioral Reviews, 2020), reviewing 31 behavioral studies, found a positive association between ADHD traits and divergent thinking — the ability to generate multiple original ideas from a single starting point. Importantly, ADHD was not associated with convergent thinking (finding a single correct solution), and stimulant medication did not negatively affect creative ability. White and Shah (2006, 2011, 2016) found that college students with ADHD scored higher on tasks requiring conceptual expansion and the ability to overcome knowledge constraints.

The contested aspect: studies using clinical ADHD samples (diagnosed individuals) show less consistent results than studies using trait-based or subclinical samples. Barkley himself has noted that confounding variables — particularly IQ — must be controlled for before creativity findings can be attributed to ADHD specifically. The scientific consensus is that divergent thinking is a plausible strength in ADHD but the evidence is not yet definitive.

### **Hyperfocus — Documented, Mechanism Debated**

The ability to sustain deep, intense concentration on tasks that are novel, interesting, or urgent — often called hyperfocus — is one of the most consistently self-reported strengths in qualitative studies of adults with ADHD. Stolte et al. (Comprehensive Psychiatry, 2024) found a positive quantitative correlation between ADHD traits and self-reported hyperfocus in a general population sample.

The mechanism is understood as dopaminergic: when a task sufficiently engages the brain's reward circuits (through interest, urgency, challenge, or novelty), dopamine levels rise enough to activate PFC function that is normally underactivated. This explains why individuals with ADHD can appear to have "no problem paying attention" to highly interesting tasks while struggling severely with routine ones — the difference is neurochemical, not motivational.

### **Additional Self-Reported Strengths — Qualitative Evidence Only**

Qualitative research consistently identifies additional self-reported strengths including: high energy and drive, empathy, cognitive flexibility, adventurousness, and humor. Stolte et al. (2024) found quantitative support for a positive association between ADHD traits and cognitive flexibility, though sociability and perseverance showed negative correlations in their sample.

A systematic review by Bisset et al. (2023) and the qualitative review by Ginapp et al. (2022) both note that, while many adults with ADHD identify positive features of the disorder, these are primarily derived from self-report, and the research base is small and methodologically limited. Most qualitative samples have overrepresented successful, highly educated adults with ADHD, making generalization to the broader ADHD population difficult.

**Summary:** Divergent thinking and hyperfocus have the strongest research support as genuine cognitive advantages associated with ADHD. Additional strengths are widely reported in lived experience research but lack the controlled, large-scale quantitative evidence that challenge findings have accumulated. This asymmetry in the evidence base is itself a significant finding — the field has studied deficits far more thoroughly than strengths.

## **Research Question 6**

**What does the research identify as effective tools, strategies, medications, and resources for improving quality of life with ADHD?**

### **Pharmacological Interventions — Strongest Evidence Base**

The Faraone consensus (2021) and the comprehensive Lancet Psychiatry meta-analysis (Cortese et al., 2025) agree that stimulant medications — specifically amphetamines and methylphenidate — are the most effective and best-evidenced interventions for ADHD:

*"Stimulants and atomoxetine, among all available pharmacological and non-pharmacological interventions for ADHD, are the only ones to be rated as efficacious in terms of ADHD core symptoms according to both self-reported and clinician-reported scales."*

A separate meta-analysis (Journal of the American Academy of Child and Adolescent Psychiatry, 2024) of 17 randomized controlled trials found amphetamines (Hedge's  $g = 0.51$ ), methylphenidate ( $g = 0.38$ ), and atomoxetine ( $g = 0.30$ ) all significantly more efficacious than placebo in improving quality of life, with moderate effect sizes. Importantly, ADHD medications were noted to reduce accidental injuries, substance abuse, educational underachievement, and other adverse outcomes — not just symptom severity.

**Critical note:** Medications are not tolerated by all individuals, evidence on long-term outcomes is limited, and they do not reliably improve quality-of-life measures beyond symptom reduction. Medication alone is consistently described in the literature as insufficient for optimal outcomes in adults.

### **Cognitive Behavioral Therapy (CBT) — Strong Evidence for Adults**

*"CBT for adults with ADHD was effective in reducing both core and emotional symptoms. Decreases in depression and anxiety were predicted by the reduction of core ADHD symptoms. An increase in self-esteem and quality of life were also observed."*

Liu et al. (2023), in a meta-analysis of 28 randomized controlled trials, found that CBT for adults with ADHD produced significant reductions in both core ADHD symptoms and co-occurring emotional symptoms. A systematic review of 53 papers (Springer, 2020) found that 92% of non-pharmacological intervention studies showed significant positive effects on either primary or secondary ADHD outcomes, with CBT showing the strongest empirical support. CBT-ADHD specifically targets time management, organization, prioritization, and cognitive restructuring of negative thoughts — the practical life-management deficits identified in Research Question 4.

## **External Structure and Environmental Design — Widely Recommended**

Across Barkley, Brown, and the Faraone consensus, a consistent recommendation is to externalize what the ADHD brain cannot reliably hold internally. This includes: written capture systems, visual timers, calendars with alerts, pre-established routines, checklists, and accountability to another person. Barkley's framework explicitly states that effective interventions must be delivered "at the point of performance" — immediately before or during the moment when the skill is needed, not in advance. This has significant design implications: reminders and prompts must be timely, visible, and require minimal retrieval effort.

## **Coaching — Evidence Emerging**

ADHD coaching — a structured, accountability-based relationship focused on daily execution rather than psychological processing — is identified as a primary recommendation by multiple expert panels in the Bisset et al. (2023) systematic review. *Frontiers in Psychology* (2022) found coaching to be among the most commonly recommended interventions by ADHD experts for adult functioning and workplace performance. Evidence from randomized controlled trials is more limited than for medication and CBT, but the expert consensus is consistently favorable, particularly for adults managing complex life demands.

## **Exercise — Documented Benefit**

Multiple meta-analyses cited in the Faraone consensus (2021) document that aerobic exercise produces measurable improvements in ADHD symptoms — particularly attention and executive function — through neurochemical mechanisms similar to stimulant medications (dopamine and norepinephrine release). The *Lancet Psychiatry* meta-analysis (2025) notes mindfulness showed large benefits at extended follow-up as the only non-pharmacological intervention to demonstrate this, though evidence was limited. Exercise is widely recommended as an adjunct to other treatments.

## **Research Question 7**

### **What do people with ADHD report as their biggest unmet need — a struggle or gap for which no adequate solution currently exists?**

This research question is answered primarily from qualitative and consumer-perspective research. The evidence base is smaller than for clinical questions, but several themes are consistently replicated across independent studies.

#### **Finding 1: Treatment That Goes Beyond Medication — Consistently Identified**

*"Consumer-identified needs related to: treatment that goes beyond medication (12 studies); improved ADHD-related education and training (17 studies); improved access to clinical services, carer support and financial assistance (14 studies)."*

Bisset et al. (Journal of Child Psychology and Psychiatry, 2023), in a systematic review of 23 peer-reviewed studies across eight countries, found that the most consistently reported unmet need among adults and caregivers with ADHD was access to multimodal care — treatment that addresses the cognitive, emotional, social, and practical dimensions of ADHD, not medication alone. Twelve of 23 studies identified this need independently.

#### **Finding 2: Practical Daily Life Support — Gap in Existing Interventions**

The qualitative review by Ginapp et al. (Frontiers in Psychiatry, 2022), synthesizing 35 qualitative studies of adults with ADHD, found that the challenges adults describe most frequently are not primarily clinical — they are practical: managing job demands, interpersonal relationships, daily routines, and the cognitive overhead of a complex adult life. The existing evidence base for interventions targeting these practical domains (particularly workplace and family functioning) was identified as severely limited by multiple expert panels in the Frontiers in Psychology systematic review (2022).

#### **Finding 3: Duration and Continuity of Care — Documented Gap**

*"Adults with ADHD, despite receiving medication, experienced impairments and challenges in many aspects of their daily life... Adults with ADHD described various unmet needs, especially those relating to the duration of treatment effect."*

Brown et al. (Primary Care Companion CNS Disorders, 2019), in qualitative interviews with 31 adults with ADHD who were already medicated, found that medication coverage did not extend adequately across the full waking day, leaving significant impairment in evening hours — precisely when family responsibilities are highest for adults with children. This is a structural gap: the current pharmacological standard of care is not designed around the demands of adult family life.

#### **Finding 4: Late Diagnosis and Self-Understanding — Unmet Need**

Multiple qualitative studies in the Ginapp et al. review document that adults diagnosed with ADHD in adulthood (as opposed to childhood) describe a specific and underserved experience: the diagnostic process provided relief and explanation, but was not typically accompanied by adequate psychoeducation, practical support, or reinterpretation of prior life history. The experience of being diagnosed late — after years of self-blame, shame, and unexplained failure — constitutes a distinct unmet need for support that is not addressed by standard clinical interventions.

#### **Finding 5: Community and Peer Connection — Identified Need**

Visser et al. (Journal of Attention Disorders, 2025), in a qualitative multi-stakeholder study, found that young adults with ADHD specifically identified peer community — connection with others who share the ADHD experience — as a highly valued and underserved resource:

*"What I think we need is more community... Finding others has been very helpful to me. I've actually learned as much, if not more, from other people that I know have ADHD... other than from my therapy and that's because you can just very practically exchange things with each other."*

**Summary:** The research consistently identifies five unmet needs: (1) multimodal care beyond medication, (2) practical daily life tools and support, (3) care that extends beyond office hours and across the full demands of adult life, (4) support for adults who received late diagnoses, and (5) peer community and shared experience. None of these needs are adequately addressed by any currently available standard clinical intervention.

## Cross-Question Synthesis

### Consensus Findings — Primary Problem Statements

The following findings represent areas where independent research institutions, large-scale meta-analyses, and clinical consensus agree. These are the most defensible statements for use as primary problem statements in design work:

1. **ADHD is a neurodevelopmental disorder, not a behavioral choice.** This is universally agreed upon across DSM-5, Barkley, Brown, Faraone, and all major clinical bodies.
2. **Executive function impairment is the central deficit.** All major researchers and the DSM-5 agree. The brain's self-management system is specifically and measurably impaired.
3. **ADHD produces a measurable and distinct relationship with time.** Time blindness is documented across Barkley's decades of research and confirmed in Faraone's consensus. Adults with ADHD do not experience time the way neurotypical adults do.
4. **The gap between knowing and doing is neurological, not motivational.** Barkley's core framework — that ADHD is a disorder of performance, not knowledge — is widely accepted. This has profound implications for intervention design.
5. **Medication is effective but insufficient.** Multiple meta-analyses confirm stimulants are the most evidence-based intervention, while also confirming they do not adequately address quality-of-life, adult functioning, or daily life management.



6. **Multimodal care is needed but not widely available or accessible.** 23 independent studies across 8 countries identify this as the primary consumer unmet need.

## Considerations — Well-Supported But Contested or Limited Evidence

- **Emotional dysregulation as a core feature:** Strong evidence in multiple peer-reviewed studies; not yet in DSM-5. Worth tracking as emerging consensus.
- **Cognitive strengths (divergent thinking, hyperfocus):** Plausible and partially supported; evidence not yet equivalent in scale or consistency to challenge findings.
- **Exercise as a meaningful adjunct intervention:** Evidence present but not yet from large-scale RCTs.
- **ADHD coaching as effective:** Expert consensus strong; controlled trial evidence limited.

## Bibliography

All sources listed are peer-reviewed or institutional primary sources. Sources are listed in order of first citation.

7. **American Psychiatric Association.** *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed. Arlington, VA: APA, 2013.
8. **Barkley, R.A.** "Attention-Deficit/Hyperactivity Disorder, Self-Regulation, and Time: Toward a More Comprehensive Theory." *Journal of Developmental and Behavioral Pediatrics* 18, no. 4 (1997): 271–79. <https://pubmed.ncbi.nlm.nih.gov/9276836/>
9. **Brown, T.E.** "ADD/ADHD and Impaired Executive Function in Clinical Practice." *Current Psychiatry Reports* 10, no. 5 (2008): 407–11. <https://pubmed.ncbi.nlm.nih.gov/18803914/>

10. **Faraone, S.V., et al.** "The World Federation of ADHD International Consensus Statement: 208 Evidence-based Conclusions about the Disorder." *Neuroscience & Biobehavioral Reviews* 128 (2021): 789–818.  
<https://doi.org/10.1016/j.neubiorev.2021.01.022>
11. **Arnsten, A.F.T.** "Neuronal Mechanisms Underlying Attention Deficit Hyperactivity Disorder: The Influence of Arousal on Prefrontal Cortical Function." *Biological Psychiatry* 69, no. 12 (2009).  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2863119/>
12. **Tripp, G., and J.R. Wickens.** "Neurobiology of ADHD." *Neuropharmacology* 57, no. 7–8 (2009): 579–89.
13. **Siqueira, I.M.V., et al.** "ADHD in Adults and Relationship with Executive Functioning and Quality of Life." *European Psychiatry* (2025).  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12438638/>
14. **Graziano, A., et al.** "Evidence of Emotion Dysregulation as a Core Symptom of Adult ADHD: A Systematic Review." *PLOS ONE* 18, no. 1 (2023): e0280131.  
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0280131>
15. **Cortese, S., et al.** "Comparative Efficacy and Acceptability of Pharmacological, Psychological, and Neurostimulatory Interventions for ADHD in Adults: A Systematic Review and Component Network Meta-Analysis." *The Lancet Psychiatry* (2025).  
[https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366\(24\)00360-2/fulltext](https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366(24)00360-2/fulltext)
16. **Liu, K.K., et al.** "Effectiveness of Cognitive Behavioural-Based Interventions for Adults with ADHD Extends Beyond Core Symptoms: A Meta-Analysis of Randomized Controlled Trials." *Psychology and Psychotherapy* 96 (2023): 543–59.  
<https://pubmed.ncbi.nlm.nih.gov/36794797/>
17. **Hoogman, M., et al.** "Creativity and ADHD: A Review of Behavioral Studies, the Effect of Psychostimulants and Neural Underpinnings." *Neuroscience & Biobehavioral Reviews* 119 (2020): 66–85.  
<https://doi.org/10.1016/j.neubiorev.2020.09.029>
18. **Stolte, M., et al.** "Associations between ADHD Traits and Self-Reported Strengths in the General Population." *Comprehensive*

*Psychiatry* 130 (2024).

<https://www.sciencedirect.com/science/article/pii/S0010440X24000129>

19. **Bisset, M., et al.** "Practitioner Review: It's Time to Bridge the Gap — Understanding the Unmet Needs of Consumers with ADHD." *Journal of Child Psychology and Psychiatry* 64, no. 7 (2023).

<https://pubmed.ncbi.nlm.nih.gov/36651107/>

20. **Ginapp, C.M., et al.** "The Lived Experiences of Adults with ADHD: A Rapid Review of Qualitative Evidence." *Frontiers in Psychiatry* 13 (2022).

<https://www.frontiersin.org/journals/psychiatry/articles/10.3389/fpsyt.2022.949321/full>

21. **Brown, T.E., et al.** "The Patient Perspective: Unmet Treatment Needs in Adults with ADHD." *Primary Care Companion CNS Disorders* 21, no. 3 (2019).

<https://www.psychiatrist.com/pcc/patient-perspective-on-unmet-needs-in-adhd-treatment/>